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Community-Based Approaches to Climate Change: Pathways for Sustainable Development—A Review of Academic and Grey Literature

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Abstract: This review explores how community-based approaches can offer effective, context-specific strategies that both adapt to and mitigate climate change, while simultaneously advancing the Sustainable Development Goals (SDGs). It interrogates how bottom-up activities have helped strengthen vulnerable populations, especially in climate-affected regions, to be more resilient and achieve a meaningful transition. A structured review of peer-reviewed and grey literature published between 2015 to 2025 was undertaken. The search identified 153 records; after duplicates were removed and titles, abstracts and full texts were screened, 21 publications met the inclusion criteria and were included in the final synthesis. The included sources comprised peer-reviewed journal articles together with 4 grey literature sources from the United Nations SDGs, the Paris Agreement and the Intergovernmental Panel on Climate Change (IPCC). The results reveal that community-based approaches are context-specific, foster social inclusion, and enhance the adaptive capacity. They align with the SDGs 1, 2, 5, 6, 7, 10, 11, 13, 14, 15, 16 and 17. Such initiatives can be scaled up, with policy enabling and partnership, as evidenced in Indonesia's Climate Village Program (Proklim). This review highlights the benefits of using indigenous knowledge in climate practices and the production of co-created knowledge integration, and multi-stakeholder governance. However, gaps in funding, poor financial monitoring, and restrictive legal systems often make it harder for local communities to adapt. This paper integrates new information related to the ability of community-based approaches to bridge the global-local divide in climate governance and offers a new approach to resilience-building based on climate justice in a participatory framework.

Keywords: Climate resilience; Community-based approaches; Local knowledge; Sustainable development; Sustainable Development Goals; Community-based adaptation

1 Introduction

Climate change is one of the greatest threats faced by mankind in the 21st century and its effects are having a huge effect on natural ecosystems, socio-economic and the developmental agendas across the world. The effects of climate change are becoming increasingly apparent in vulnerable regions of the Global South, as a result of a growing number of natural disasters, erratic weather patterns, and rising sea levels. The Global South, while contributing the least to greenhouse gas emissions, bears a disproportionate share of the impacts, including food insecurity, loss of livelihoods, and environmental degradation [1].

Conventional top-down approaches facilitated by national governments and global organizations have affected highly important frameworks, including the Paris Agreement [2] and the United Nations Sustainable Development Goals (SDGs) [3], but its implementation remains low, particularly in terms of ensuring climate adaptation at the grassroots level. In this regard, one comes across the important role of community-based approaches to climate change. Community-based approaches focus on local communities, indigenous or local knowledge, and the empowerment of communities to plan and implement responses to climate issues that are unique to their societal or cultural setting and the ecosystem of which they form part of [4].

Community-driven climate adaptation and mitigation strategies reflect the core values of sustainable development. Amongst other measures, engaging in climate-resilient agricultural practices, decentralized renewable energy

solutions, water conservation, disaster risk reduction (DRR) and community forestry, community-based approaches can address several SDGs together. These include SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 6 (Clean Water and Sanitation), SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 15 (Life on Land). It also follow the SDG 17 (Partnerships for the Goals), by example through cooperation with local communities, civil society organizations, governmental levels, and research organizations [5]. Although community-based initiatives are good options in making climate-resilient developments, it does not come without its problems. Poor funding and institutionalization, poor scalability, and failure to be integrated into official formal governance systems are some of the issues that inhibit their possible effectiveness. However, an increasing appreciation is that locally run initiatives offer context sensitive, cost effective and sustainable substitutes or additions to national and international climate action plans. Community-based initiatives are emerging as promising approaches to lessen the impact of climate change while empowering people and bolstering community resilience [6]. The study [4] indicated that a growing number of community initiatives at the grassroots level are emerging and are actively responding to challenges faced by climate change. These bottom up processes require the community to be active agents in the process and uses their local knowledge to implement and develop context specific solutions positioned with climate goals. The study [7] opined that community-led interventions can differ widely. Some interventions are linked to indigenous knowledge with the awareness that the community and its resources, require help externally in terms of knowledge and/or resources to address the risks beyond the capacity of the community. Although numerous studies have examined individual community-based adaptation initiatives, there remains limited synthesis of how community-based approaches collectively contribute to climate adaptation, mitigation, and sustainable development across different geographical contexts [8–10]. This review addresses this gap by integrating current academic and grey literature published between 2015 and 2025.

This review examines the scholarly literature on community-based approaches to climate change, particularly on the role these strategies play as a pathway to sustainable development. It explored the theoretical basis of community-based approaches, their implementation through different regions and sectors and the compatibility of community-based approaches with agendas of sustainable development globally. However, by exploring case studies and other empirical data, the review seeks to both highlight the transformative potential of community-based approaches and reveal their limitations in addressing the complex interplay between climate vulnerability and sustainable development.

2 Literature Review

2.1 Defining Community-Based Approaches in Relation to Climate Action

Community-based approaches to climate change are local, bottom-top strategies, which seek to contribute to the capacity of adaptivity and resilience of vulnerable populations. In contrast to conventional top-down, climate policies, it aims at empowering the local actors, especially those living in marginalized or climate-sensitive areas, to design, carry out and monitor solutions on the basis of lived experience and local knowledge [8]. These methods are founded on the concept of participatory development by which communities are not recipients of foreign assistance, but instead they are active people who are able to innovate and guide sustainable changes. The study [9] highlighted that, over the years, community-based approaches have increasingly become central to global discussions on climate issues. They foster ownership and encourage culturally appropriate actions, allowing decisions to be made within the local context. By their nature, they are interdisciplinary, due to having combined elements of environmental governance, social justice, and DRR. Notably, they are consistent with the perspective of climate justice, which aims to correct systematic inequities, which place certain sections of society at a higher risk of being affected by environmental shock [4]. The SDGs, especially SDG 13 (Climate Action), SDG 11 (Sustainable Cities and Community), and SDG 17 (Partnerships for the Goals), overlap with community-based approaches too. Such strategies are implemented through active community participation and they not only enhance climate resilience, but also lead to greater development goals, such as poverty reduction, gender equality and livelihoods. Their wide-ranging relevance has positioned community-based approaches to climate change as a crucial means of delivering local responses alongside national and global strategies.

2.2 Community-Based Adaptation Strategies

One of the most researched topics about community-based approaches in the literature, is community-based adaptation. It is defined as a process through which communities determine, evaluate, and respond to vulnerabilities associated with climate by prioritizing participatory risk assessment, sharing of knowledge, as well as low-cost and scalable responses. These could be climate-smart farming practices, rain-catchment structures, preliminary notification frameworks, and early warning systems [11]. The study [11] presented community-based adaptation strategies by considering adaptive decision-making by implementing indigenous knowledge into scientific data. This combination has been crucial in solving agricultural problems in many rural Africa and South Asia communities, which are characterized by unpredictable rainfall and climatic variations. In Ethiopia, community-based soil

and water conservation initiatives at the Tigray region have seen farmers experience increased vegetation cover, strengthening the natural resource base and supporting rural livelihoods [6]. According to the study [12], in India, West Bengal have decreased their dependence on climate-sensitive and rain-fed farming occupations through livelihood diversification, which has contributed to building long-term resilience. Notably, community-based adaptation depends on a case. Interventions have been observed to largely depend on factors like cultural norms, gender roles, as well as available institutional capacity. This highlights the need for adaptive programming and flexible policy approaches that consider community preferences and capacities.

Scholarly literature [7] has therefore pointed out that effective community-based adaptations must have the following:

- Adequate financial and external support
- Strong linkages between local and higher-level institutions
- Strengthening local adaptive capacity
- Participatory planning, implementation and learning

These factors enable community participation to translate into long-term climate resilience and meaningful social change.

2.3 Participatory Action and Co-Production of Knowledge

The key feature of community-based approaches involves participation, especially in building community ownership, trust and sustainability over the long term. The methodologies of Participatory Action Research (PAR) are popular at a global level in co-developing interventions with the members of the community, so that interventions could be found that represent real priorities and practical needs [13]. Knowledge is co-produced when external expertise is combined with local communities in designing, testing and implementing solutions, which is very much applicable in climate forecasting and adaptation planning [5]. As indicated in the study [14], co-produced knowledge supports iterative learning, increases credibility and enhances adaptive capacity. An example is demonstrated how farmers in Lushoto District, Tanzania, integrated Indigenous weather forecasting with scientific seasonal forecasts from the Tanzania Meteorological Agency through collaborative knowledge-sharing groups, enabling farmers to access more locally relevant climate information to inform agricultural planning. Participation that is empowerment-based may be more likely to have a transformative effect and communities have the decision-making ability as well as control over resources [13]. There should also be gender-sensitive approaches. The inclusion of women in making climate decisions is important because of their roles in food systems, water management and household resilience. Women leadership is more sustainable and more likely to be preserved through programs. As an example, the study [4] discovered that in Brazil, the collaborative water management of the cooperatives of women in the semi-arid regions reduced the risk of drought events due to collective water governance and agroecological approaches in the community.

2.4 Examples of Community-Led Initiatives and Practical Outcomes

Table 1 represents a synthesis of selected studies examining community-based responses to climate change across different geographical contexts. The synthesis highlights the climate-related challenges experienced by communities, the actions undertaken at community level, their alignment with the SDGs, and the outcomes reported in the literature. Collectively, the studies demonstrate the importance of community participation, local knowledge, collective action and locally appropriate strategies in addressing climate-related vulnerabilities and promoting sustainable development [15].

Table 1. Summary of studies on community-based climate change responses

Location	Climate Issue	Community Action	Related SDGs	Reported Outcome
Indonesia	Climate change and increasing climate-related risks, including impacts on water resources, livelihoods, rural economies and disaster vulnerability.	Community-based micro-watershed management, involving local communities in sustainable natural-resource management, livelihood diversification, climate-resilient infrastructure and disaster-risk reduction. The approach also emphasises empowering local communities, multi-stakeholder partnerships and integrating climate resilience into village planning and policy.	SDG 1, SDG 2, SDG 6, SDG 11, SDG 13	Participatory micro-watershed management; sustainable natural-resource management; livelihood diversification; climate-resilient infrastructure; disaster-risk reduction; community empowerment and partnerships [5]

Location	Climate Issue	Community Action	Related SDGs	Reported Outcome
Zimbabwe	Drought, food/water scarcity, habitat disruption, biodiversity loss and human–wildlife conflict affecting Zimbabwe's tourism sector.	Community-centred conservation, local participation in wildlife management, tourism revenue-sharing, water conservation, sustainable land management and protection of wildlife corridors.	SDG 6, SDG 8, SDG 13, SDG 15	The study found significant relationships between all six climate-related factors and sustainable tourism. It recommends integrated climate adaptation, biodiversity conservation and community participation to strengthen tourism resilience. [15]
India	Increasing exposure to climate-induced extreme events, particularly cyclones, storm surges, flooding and coastal hazards, which threaten agricultural production, fishing activities and household livelihoods.	Households diversified their livelihoods by shifting from traditional nature-based activities, particularly agriculture and fishing, towards non-farm income-generating activities.	SDG 1, SDG 2, SDG 8, SDG 13	Livelihood diversification provided alternative income opportunities and was identified as a strategy for addressing climate-induced poverty. The study also found that food security explained 81.7% of regional livelihood diversification in its regression model [12]
Ethiopia	Climate change vulnerability, particularly among poor and marginalised communities; climate risks interacting with existing poverty and development challenges.	Community-led adaptation; use of local knowledge and existing adaptive capacities; participatory decision-making; communities identify priorities and make adaptation choices; governments/NGOs provide facilitation and support rather than imposing solutions.	SDG 1, SDG 5, SDG 10, SDG 11, SDG 13, SDG 16, SDG 17	Community-based adaptation can strengthen local adaptive capacity and empower communities, while making adaptation more responsive to local priorities. However, the authors identify challenges including tokenistic participation, unequal power relations, limited finance/institutional support, and difficulties integrating community-based adaptation into wider development planning. [6]
Zambia	Energy poverty and limited access to clean, sustainable energy	Communities were engaged in solar PV adoption through peer learning, observational learning, local networks and consultation. Residents observed neighbours using solar technologies and sharing their experiences. The study recommends community-based demonstrations involving trusted local champions, stronger consultation and greater community ownership rather than externally imposed interventions.	SDG 1, SDG 7, SDG 8, SDG 10, SDG 11, SDG 13	Observational learning and peer influence increased confidence and willingness to adopt and pay for solar PV [16]. Seeing functioning solar systems and their practical benefits encouraged other households to consider adoption. The authors conclude that community ownership, effective information-sharing and local participation are important for sustaining solar-energy initiatives.
Afghanistan	Rising temperatures, erratic rainfall, prolonged droughts, water scarcity and extreme weather events are affecting agricultural production, food security, water availability and rural livelihoods.	The article recommends community-based climate adaptation, including climate-smart agriculture, water conservation and management, renewable energy, integration of traditional/local knowledge with scientific knowledge, and participatory research in which communities help co-create locally appropriate solutions.	SDG 2, SDG 6, SDG 7, SDG 8, SDG 13, SDG 15	The review finds that climate change is reducing agricultural productivity, increasing water and food insecurity, and worsening rural vulnerability and poverty. It identifies community participation, climate-smart agriculture, improved water management, renewable energy and stronger institutional support as pathways for building rural resilience and sustainable livelihoods [17].

Note: SDG = Sustainable Development Goal.

3 Methodology

This study adopted a structured literature review approach to examine the contribution of community-based approaches to climate change adaptation and mitigation as pathways towards sustainable development. A structured literature review provides a systematic yet flexible means of identifying, evaluating, and synthesizing existing knowledge from diverse sources to develop a comprehensive understanding of an emerging field. This approach was considered appropriate because community-based climate action spans multiple disciplines, including environmental science, sustainable development, public policy, DRR, and community development.

3.1 Literature Search Strategy

A comprehensive search of the literature was undertaken using electronic databases including Scopus, and ScienceDirect. ResearchGate and Google Scholar was not used as a database but was consulted to locate full texts and grey literature. To complement the peer-reviewed literature, relevant grey literature was also consulted separately with four grey literature sources to accommodate relevant international policy frameworks such as the United Nations SDGs, the Paris Agreement and Intergovernmental Panel on Climate Change (IPCC) reports.

The search strategy combined keywords related to community-based approaches, climate change adaptation and sustainable development using the Boolean operators AND and OR. The search was limited to English-

language publications published between 2015 and 2025. The following search string was applied: TITLE-ABS-KEY((“community-based approach*” OR “community-based adaptation” OR “community-led adaptation” OR “grassroots climate action” OR “participatory adaptation”) AND (“climate change adaptation” OR “climate resilience” OR “climate action”) AND (“sustainable development” OR “sustainable livelihood*”)) AND PUBYEAR > 2014 AND PUBYEAR < 2026 AND LIMIT-TO(LANGUAGE, “English”).

A total of 153 records were retrieved from Scopus and ScienceDirect and exported to EndNote for reference management. Thirty duplicate records were identified automatically based on title, author and publication year and subsequently verified manually. Following duplicate removal, 123 records were screened by title and abstract against the predefined inclusion criteria, of which 79 were excluded. The remaining 44 articles were retrieved and assessed for eligibility through full-text screening. Of these, 23 articles were excluded because they did not sufficiently address community-based approaches to climate change or their relationship to sustainable development. Consequently, 21 academic articles met the inclusion criteria and were included in the review. In addition, 4 relevant grey literature documents were identified and included to provide policy context and support the review’s sustainable-development focus. No supplementary records were identified through additional searching. The final synthesis comprised 21 academic articles and 4 grey literature documents.

3.2 Literature Selection

The review focused primarily on literature published between 2015 and 2025 to capture both foundational and contemporary scholarship on community-based approaches to climate change. Priority was given to peer-reviewed journal articles, while relevant grey literature was included where it provided important policy, conceptual, or practical insights. Publications were selected based on their relevance to community-led climate adaptation, participatory climate governance, indigenous and local knowledge systems, and their contribution to sustainable development.

Studies that focused exclusively on national climate policy without a community dimension or were not directly relevant to the review objectives were excluded from detailed analysis.

3.3 Data Synthesis

The selected literature was analyzed using a thematic synthesis approach. Key concepts, recurring themes, implementation strategies, enabling factors, barriers, and contributions to sustainable development were identified and compared across the literature. Particular attention was given to evidence relating to community participation, climate resilience, indigenous knowledge, governance, digital innovations, and alignment with the SDGs. The findings were then synthesized narratively to identify common patterns, emerging trends, and knowledge gaps.

3.4 Limitations of Methodology

Although this structured literature review provides a comprehensive synthesis of current knowledge on community-based approaches to climate change and sustainable development, several limitations should be acknowledged. First, the review relied primarily on literature published in English, which may have excluded relevant studies published in other languages, particularly from regions where community-based climate initiatives are widely implemented. Second, while multiple academic databases and selected grey literature sources were consulted, it is possible that some relevant publications were not identified because of differences in database indexing or the use of alternative terminology.

Furthermore, the review focused on synthesizing published evidence rather than conducting a quantitative assessment of study outcomes. Consequently, the findings represent a narrative interpretation of the literature rather than a statistical evaluation of the effectiveness of community-based approaches. The diversity of study designs, geographical contexts, and intervention types also limited direct comparison across studies. Finally, because climate adaptation and sustainable development continue to evolve rapidly, newly published research may provide additional evidence that was not available at the time this review was conducted. Despite these limitations, the review offers a comprehensive overview of current scholarship and identifies key trends, opportunities, and challenges that can inform future research, policy, and practice.

4 Discussion

The literature review covers a wide geographical setting, including broad regions of South Asia and Sub-Saharan Africa, offering an extensive outlook on effectiveness and barriers of community-based approaches as well as its integration into the larger developmental framework.

One of the themes that stand out clearly in the literature is the importance of strong community participation in the climate resilience. Community-based approaches to climate change engage the local residents in the process of planning, implementing, and monitoring proved to be more effective and sustainable. According to the study [8], active approaches like community mapping and knowledge co-production have the benefits of establishing ownership, trust, as well as culturally suitable adaptation solutions. In the same line of thought, the study [18] observed that

effective community-based interventions to East African vector-borne disease situations could be attributed to a factor of increasing local voices and health beliefs as part of intervention strategies. The authors [19] also insist that participatory models play a crucial role in addressing the issue of power asymmetry and involving the vulnerable population, especially women and Indigenous peoples, in a meaningful way. Nevertheless, according to the study [20], there is a gap between the intent and practice of participation, with some instances in which communities were consulted but did not have the authority of increased decision-making. Most of the literature reviewed, reports on the efficacy of the nature-based solutions into the community-led adaptation models. The study [21] showed strong evidence in Bangladesh that by using mangrove restoration and floating agriculture and re-excavating ponds, not only did the local communities increase their climate resilience, their food security, and biodiversity maintenance were also achieved. Besides alleviating flood and salinity susceptibility, these interventions helped achieve several SDGs, such as SDG 13 (Climate Action), SDG 2 (Zero Hunger), and SDG 15 (Life on Land). This is supported by the authors [20], who posit that sustainable fisheries and marine conservation efforts initiated by coastal communities in tropical countries exhibit twofold impacts of both livelihoods and the well-being of the ecosystem. In comparison, the study [22] explained that community participation in the process of biodiversity finance, including payment of ecosystem services (PES), results in the enhancement of ecological and economic outcomes.

Although traditional knowledge will be most certainly a key component of most community-based approaches, there is an emphasis on the new role of digital tools and geospatial technologies in community capacity building by multiple studies [6]. The study [23] emphasized artificial intelligence and geospatial mapping as a tool that can be used to enhance disaster preparedness and resource distribution in climate-threatened areas. The idea of community-based monitoring (CBM), especially in Indigenous communities, has additionally become eminent as data sovereignty and resilience creation [17]. These tools also enhance responsiveness of local governance as well as enabling integration to national data systems. In their work, the authors [13] further explained the utilization of smart grids and digital twins in renewable-based community utility to minimize the environmental footprints and achieve independence in energy consumption. The use of these technologies is, however, uneven, and the study [24] argued that remote locations may experience restricted access to such technology, particularly in Sub-Saharan Africa, due to infrastructural and digital divides. However, even though there have been myriads of success stories, the literature has continued to implode on various obstacles hindering both scalability and sustainability of community-based adaptations. Gaps in funding and dependence on outside donors are issues that keep appearing [20, 25]. According to the ideas expressed by the authors [26], without seeing institutional integration, community-based approaches risk being pilot projects that never have a powerful, long-lasting effect. The study [25] reported that in Afghanistan the inefficiencies of the support systems and poor governance devalues the attempts made by the villagers to adapt to the circumstances. Moreover, the lack of effective monitoring and evaluation systems prevents the possibility of evaluating and comparing the effectiveness of community-inspired interventions [26]. The study [27] found that certain forms of community-based responses can unintentionally reinforce social inequalities, as local frameworks may be shaped by elite capture, exclusionary practices, or entrenched power dynamics.

One of the aspects of this review which can be regarded as positive, is the fact that the community-based approaches have been shown to be in line with several SDGs. Community-based approaches helps in achieving gender equity, sustainable water and sanitation as well as poverty alleviation in addition to climate action. As an example, the study [28] highlighted the importance of community-based waste management projects in developing circular economy goals and making European cities more sustainable. The study [14] explained the difference between community trail planning, which is meant to facilitate eco-tourism and the preservation of green spaces-associating land use planning not only with economic resiliency but also with environmental conservation. Moreover, the study [29] highlighted the pressing need for bibliometric data that illustrates the increasing connection between climate mitigation strategies and global development discussions. Integrating scientific, indigenous, and policy frameworks into the design of community-based approaches strengthens their local relevance and broadens their reach, opening up possibilities for more holistic development planning.

The literature reviewed points to four factors that distinguish initiatives that continue beyond their initial project period from those that do not. This includes funding (a) funding arrangements, (b) local decision-making power, (c) links with government and (d) monitoring which will be discussed below.

Funding arrangements: Where funding is episodic and externally driven, sustainability mechanisms frequently fail to function after closure, leaving communities to shoulder the burden of reviving interventions [20, 26]. Evidence from the assessment [30] of 32 community-based adaptation initiatives across 20 rural communities in the Pacific Islands indicates that locally funded and implemented initiatives were more successful than those reliant on external international funding. The authors further identified the sustainability of initiatives beyond the project funding period as a significant challenge. This aligns with ex-post assessments of internationally funded adaptation projects, where mechanisms envisioned by implementers had not functioned after the project concluded [28].

Local decision-making power: Participation that stops at consultation does not sustain outcomes [20]. Where

communities hold genuine decision-making authority and control over resources, interventions are more likely to be transformative [27], and locally led decision-making is identified as a core driver of appropriate grassroots adaptation [31].

Links with government: In the absence of institutional integration, community-based adaptations risk remaining pilot projects without lasting effect [26], while enabling governance and policy coherence are what allow approaches such as PROKLIM to be institutionalised at scale [5]. By contrast, where support systems are inefficient and governance weak, community adaptation efforts are undermined [29].

Monitoring: The absence of effective monitoring and evaluation prevents comparison of outcomes across interventions [30], and where monitoring is weak, failures go unreported while success stories are promoted for scaling [28]. Effective monitoring systems are therefore a precondition for both accountability and learning.

Across all four factors, the pattern is consistent. Continuation depends less on the technical design of an intervention than on whether authority, resources and accountability are retained locally after external support ends.

5 Conclusions

The current literature review has examined the rapidly expanding body of scientific literature on the role of community-based approaches to climate change. These approaches are highlighted as critical support measures for strengthening both local resilience and sustainability. The results demonstrate that community-based approaches do not merely represent viable mechanisms to deal with the issue of vulnerabilities to climatic changes, but also very effective devices of social empowerment, participatory governance, and environmental stewardship. One of the common themes observed in the studies reviewed is the need for an inclusive and participatory processes. Climate strategies that are meaningfully implemented, monitored, and designed by the community is more contextually appropriate, trusted, and sustainable. The strength of the community driven approaches also proves to be able to integrate local knowledge systems, flexibility with the social-cultural environments and long-term commitment. Such features tend to be more effective than climate interventions that are external, and top-down. The combination of nature-based and ecosystem-based solutions in community-based approaches to climate change have found to be a way to provide broad-based benefits. These include mangrove restoration and agroecology, community water management, renewable energy microgrids, as well as more innovative interventions such as water management, and community social cohesion, which have demonstrated their ability to simultaneously increase environmental sustainability, whilst also positively transforming local livelihoods.

Lack of supportive policies and well-established monitoring mechanisms in most scenarios weakens community-based initiatives in the long term and even in cases of upscaling it. Moreover, although digital means have extended accessibility and increased accuracy of certain community-based approaches, the presence of digital and other infrastructural gaps limit access in other areas too far, and those with insufficient resources. In spite of these impediments, community-based approaches to climate change have proven to be in line with the various SDGs, including goals related to climate action, poverty reduction, gender equity, and water access. This is because they can be used as local driven engines to sustainable development. The policies to be followed to realize this potential include more emphasis on policy combination, sustainable financing mechanisms, capacity building and non-discriminatory governance in future. Research protocols that are intensive and longitudinal in nature, and assess the sustainability and generalizability of community-based approaches in varied settings are also necessary. Finally, the building of resilient societies necessitates the empowerment of communities as the frontier in taking charge in the climate action process.

Author Contributions

Conceptualization, Y.G. and R.B.; formal analysis, Y.G.; writing—original draft preparation, Y.G.; writing—review and editing, R.B.; supervision, R.B. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI and AI-assisted technologies were used during the preparation of this manuscript to assist with grammar, language refinement and sentence construction. These tools were used solely to improve the clarity, readability and presentation of the written text. They were not used to generate research findings, analyse data, or develop the scholarly arguments and conclusions of the study.

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